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5. (a) The symbol equations below show three reactions that produce salts.

(i) Complete the symbol equation for each reaction. [3]

A $Zn(s) + 2HNO_3(aq) \longrightarrow Zn(NO_3)_2(aq) + \text{..............................}(g)$

B $MgO(s) + 2\text{..............................}(aq) \longrightarrow MgCl_2(aq) + H_2O(l)$

C $Pb(NO_3)_2(aq) + 2NaCl(aq) \longrightarrow \text{..............................}(s) + 2NaNO_3(aq)$

(ii) Give the **letter**, **A**, **B** or **C**, of the reaction that forms a precipitate. [1]

....

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(b) (i) The equation below shows the reaction between dilute hydrochloric acid and sodium hydroxide solution.

$$HCl \quad + \quad NaOH \longrightarrow NaCl \quad + \quad H_2O$$

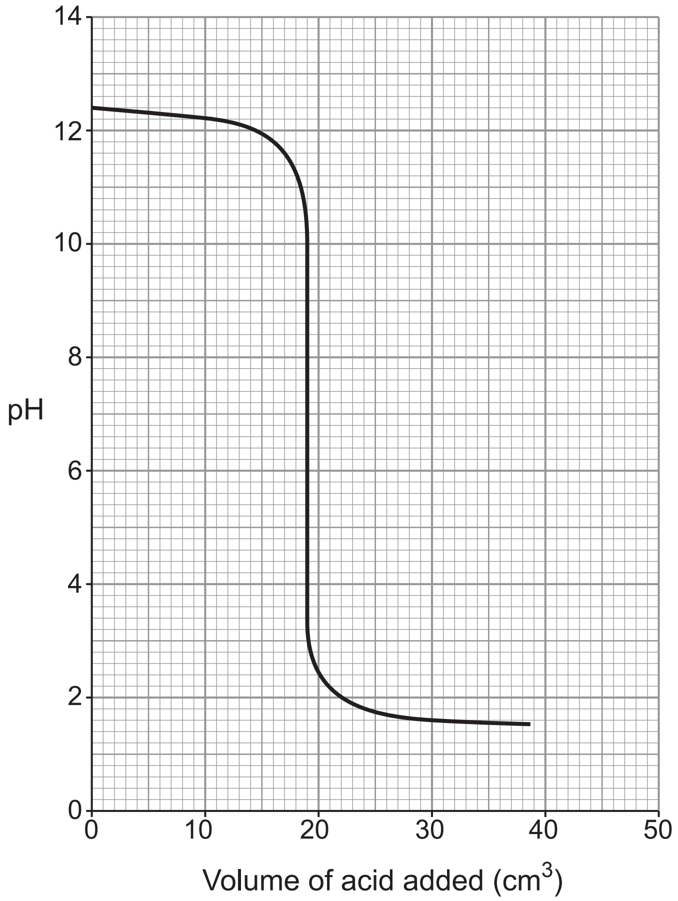
Give the term used to describe the reaction between an acid and an alkali to form a salt and water. [1]

....

(ii) Eleanor carried out an investigation to find how the pH changes when dilute hydrochloric acid is added to sodium hydroxide solution.

She gradually added dilute hydrochloric acid to 25.0 cm^3 of sodium hydroxide solution.

The graph shows how the pH of the reaction mixture changed **as the acid was added**.



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Use the graph to answer parts I–III.

I. Sodium hydroxide is an alkali. Give the pH of the sodium hydroxide solution before any acid was added. [1]

....

II. Give the volume of acid needed to completely react with the sodium hydroxide solution. [1]

.... cm^3

III. Tick (✓) the box next to the statement that correctly describes the concentrations of the acid and the alkali used in this investigation. [1]

the acid and the alkali have the same concentration ☐

the acid has a lower concentration than the alkali ☐

the concentration of the acid is half of the concentration of the alkali ☐

the acid has a higher concentration than the alkali ☐

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(iii) Eleanor measured the pH using a pH sensor. Freddie did a similar experiment but used universal indicator solution to measure the pH.

The table shows the colours of universal indicator at different pH values.

Colour	red	orange	yellow	green	blue	navy blue	purple
pH	0–2	3–4	5–6	7	8–9	10–12	13–14

Suggest why the data Eleanor collected could be more useful than Freddie's data. [1]

....

....

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